

CS4226

Internet Architecture



Tutorial 2

Question 1

Suppose the arrivals of A2 buses follow a Poisson process with rate λ . What is the probability that two consecutive arrivals come within a time period of length t ?

Question 1

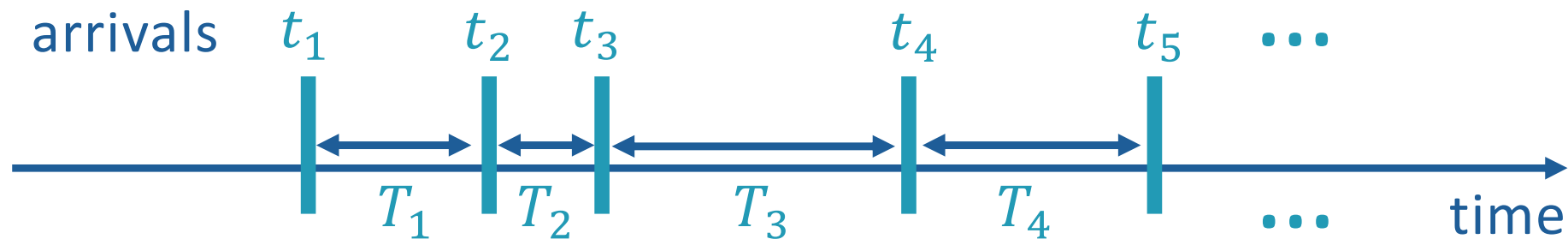
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What is a Poisson process with rate λ ?

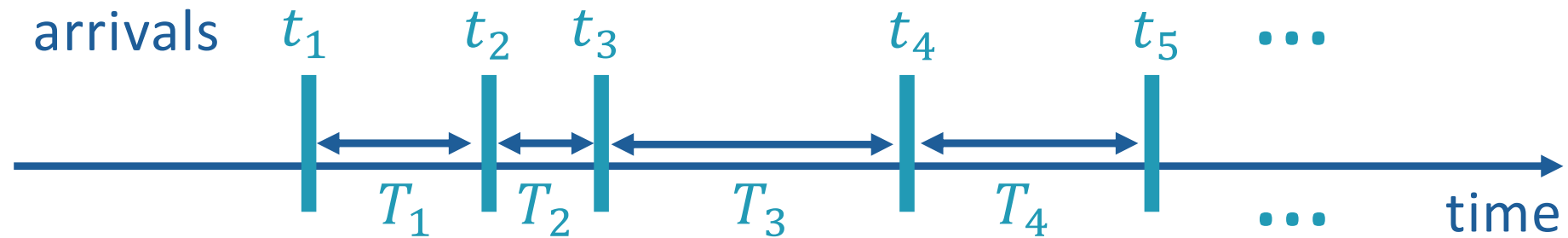
Question 1

What is a Poisson process with rate λ ?

- Events (arrivals of A2 buses) occur stochastically.
- Inter-arrival times (T_i s) are *independently and identically distributed (i.i.d.) random variables (r.v.s)*.
- Inter-arrival times (T_i s) follow an *exponential distribution* with rate λ .



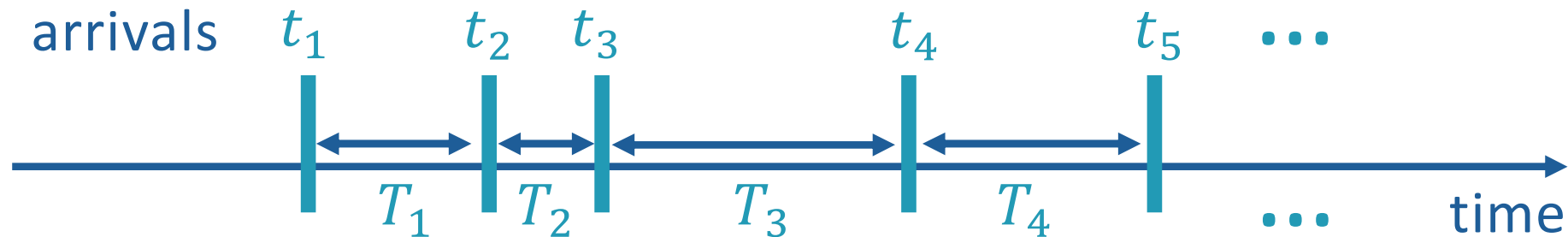
Question 1



Owing to the exponential distribution, the prob. that T_i is not longer than t is

$$P\{T_i \leq t\} = 1 - e^{-\lambda t}$$

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i.e., the prob. that two consecutive arrivals come within a time period of length t is

$$F_T(t) := P\{T \leq t\} = 1 - e^{-\lambda t}$$

Question 2

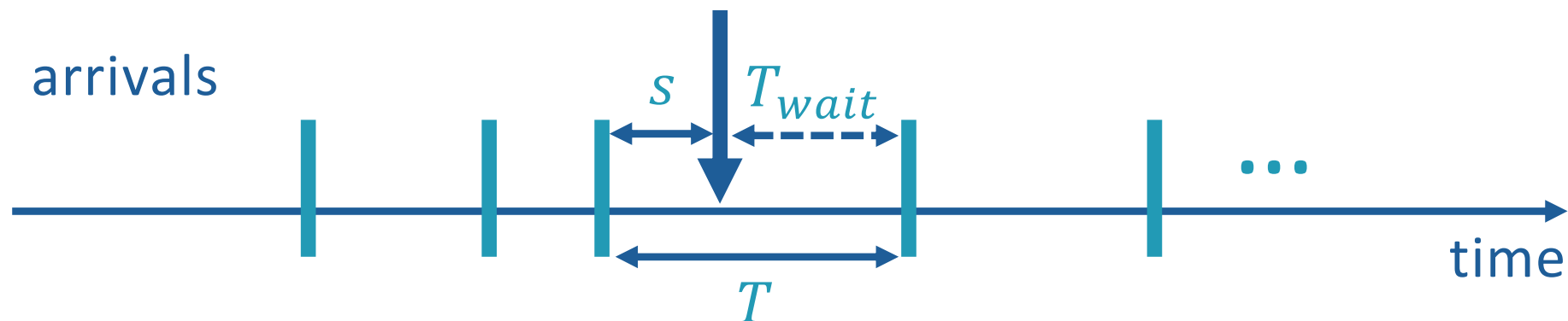
Suppose the arrivals of A2 buses follow a Poisson process with rate λ . When you arrive at the bus stop, you are told that the last A2 bus has departed for s amount of time.

What is the probability that your waiting time for the next A2 bus is longer than t ?

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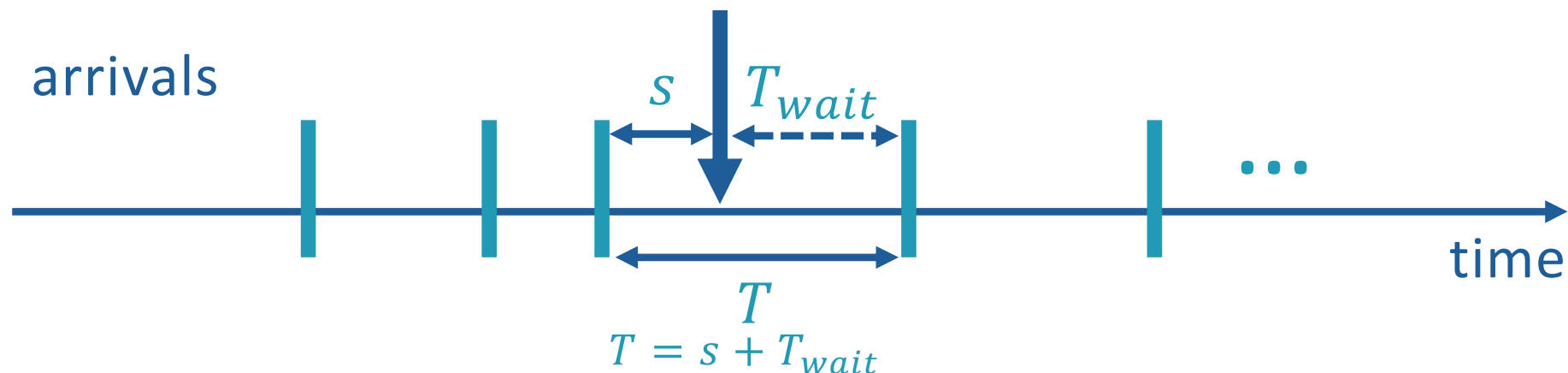
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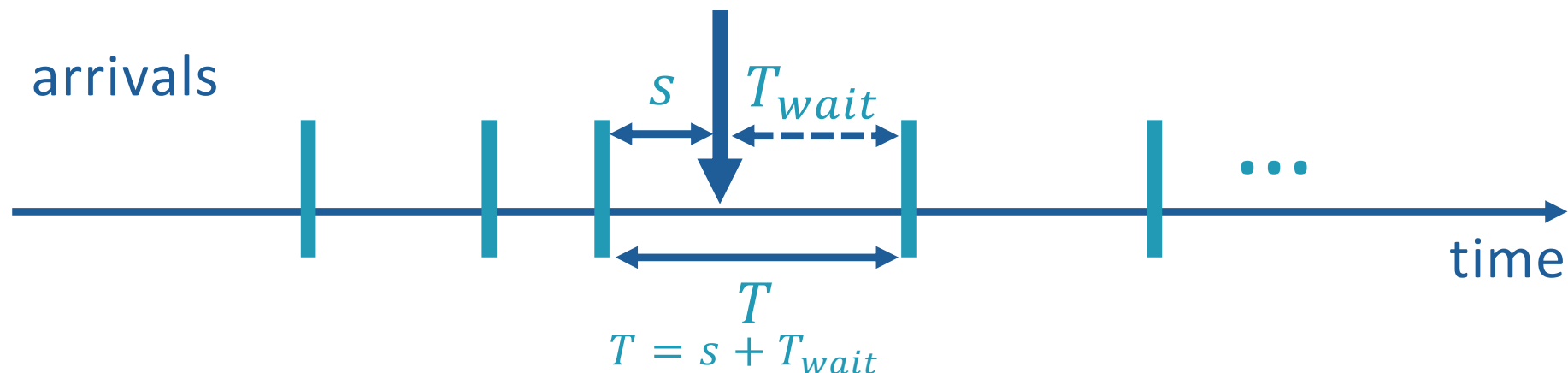
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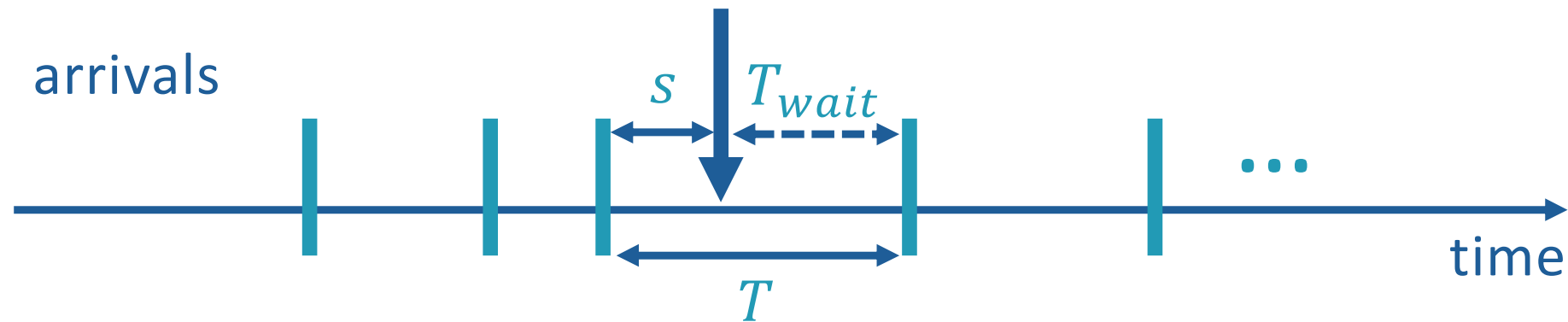
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What about T_{wait} if $T = s + T_{wait}$?

Question 2

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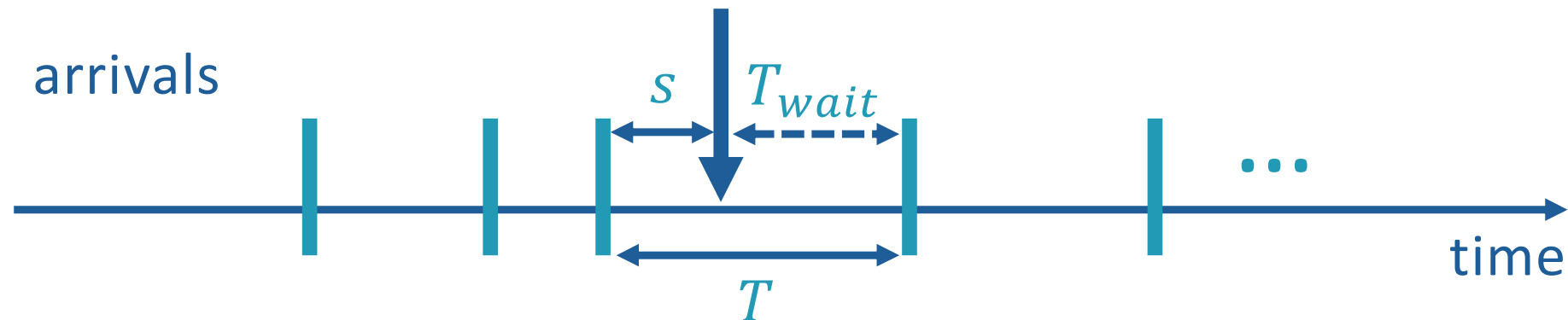


The implied condition is $T > s$, and we want to derive

$$P\{T_{wait} > t \mid T > s\}$$

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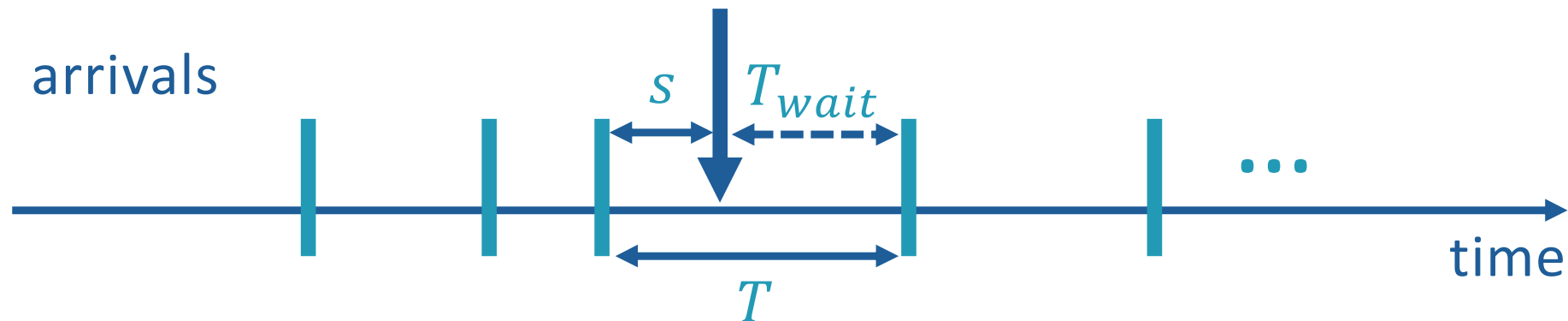
The implied condition is $T > s$, and we want to derive

$$\begin{aligned} &P\{T_{wait} > t \mid T > s\} \\ &= P\{s + T_{wait} > s + t \mid T > s\} \\ &= P\{T > s + t \mid T > s\} \end{aligned}$$

Now it is only related to the variable T .

Question 2

What about T_{wait} if $T = s + T_{wait}$?



The implied condition is $T > s$, and we want to derive

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Now it is only related to the variable T .

We already know $P\{T > t\} = 1 - P\{T \leq t\} = e^{-\lambda t}$.

How to derive the conditional prob. $P\{T > s + t \mid T > s\}$?

Question 2

How to derive the conditional prob. $P\{T > s + t \mid T > s\}$?

$$P\{T > s + t \mid T > s\} = \frac{P\{T > s + t, T > s\}}{P\{T > s\}}$$

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How to derive the conditional prob. $P\{T > s + t \mid T > s\}$?

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$$\begin{aligned} P\{T > s + t \mid T > s\} &= \frac{P\{T > s + t, T > s\}}{P\{T > s\}} \\ &= \frac{P\{T > s + t\}}{P\{T > s\}} \\ &= \frac{1 - P\{T \leq s + t\}}{1 - P\{T \leq s\}} \\ &= \frac{1 - (1 - e^{-\lambda(s+t)})}{1 - (1 - e^{-\lambda s})} = e^{-\lambda t} \end{aligned}$$

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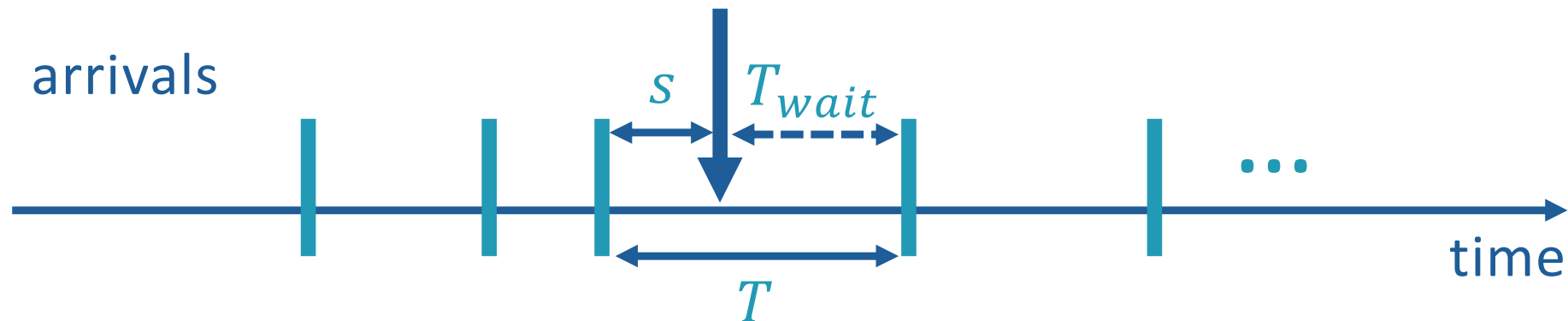
We can notice that $P\{T > s + t \mid T > s\} = P\{T > t\}$.

→ The conditional prob. is independent of s .

→ Memoryless property of Poisson process

Question 2

Memoryless property of Poisson process



Distribution of the time until next arrival (T_{wait}) is independent of the time since last arrival (s).

$$\rightarrow P\{T > s + t \mid T > s\} = P\{T > t\}$$

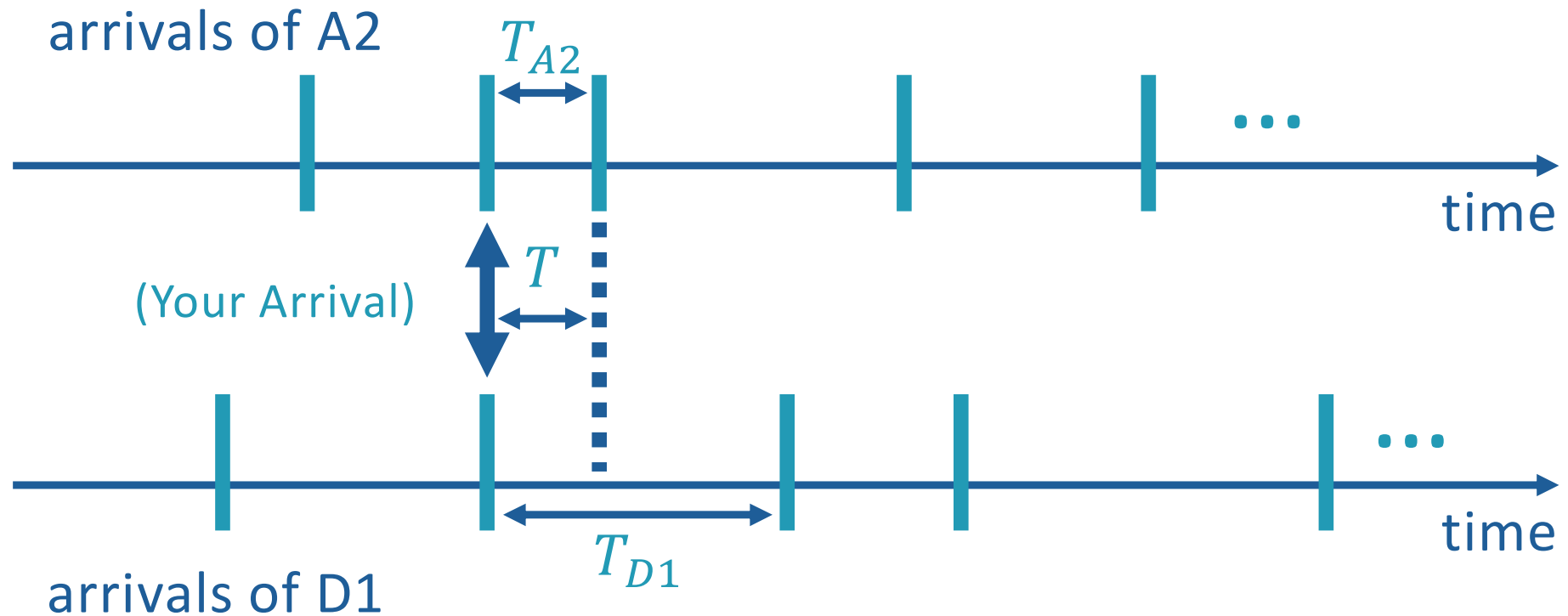
Question 3

Suppose the arrivals of A2 buses and D1 buses follow Poisson processes with rate λ_{A2} and λ_{D1} . Suppose you want to go to the Central Library from SoC and can take either A2 or D1.

Unfortunately, when you arrived at the bus stop, you see an A2 bus and a D1 bus have just departed.

How long is your average waiting time for you to get on your bus?

Question 3

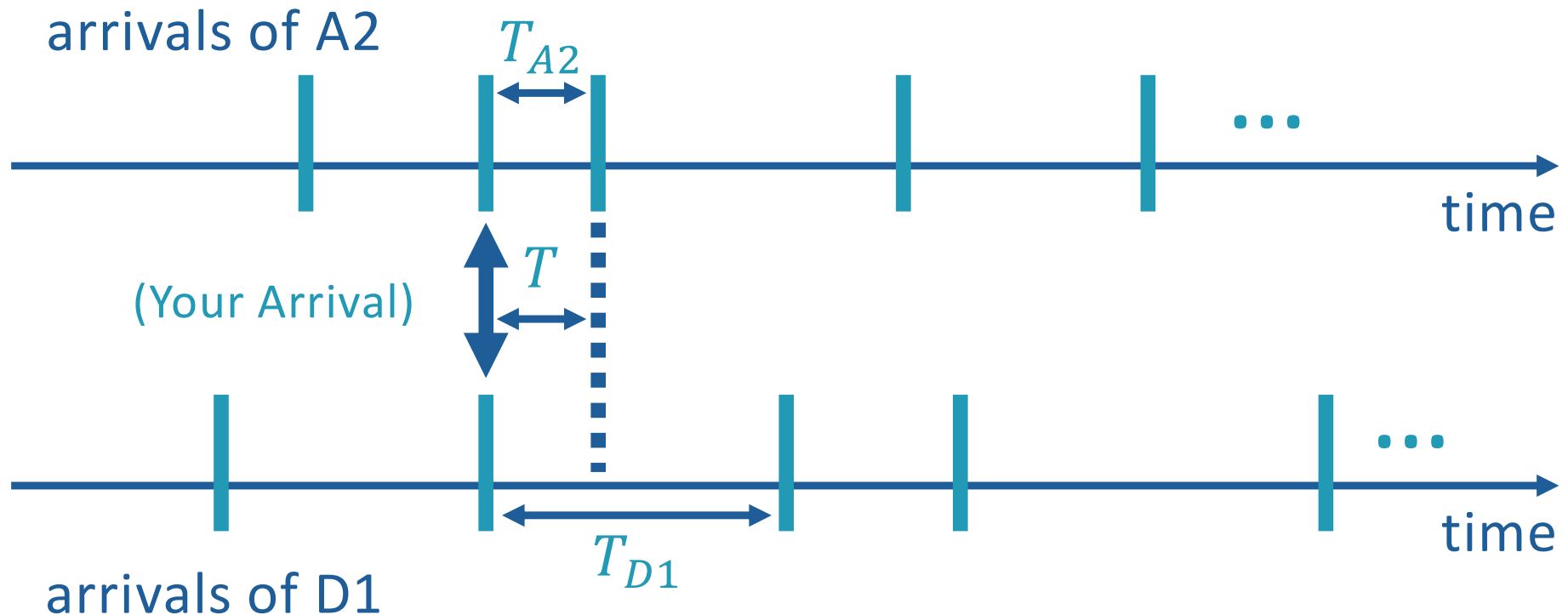


T_{A2} : Inter-arrival time of A2.

T_{D1} : Inter-arrival time of D1.

T : Waiting time for the next bus.

Question 3



T_{A2} : Inter-arrival time of A2.

T_{D1} : Inter-arrival time of D1. $\rightarrow T = \min(T_{A2}, T_{D1})$

T : Waiting time for the next bus.

$$\rightarrow P\{T > t\} = P\{\min(T_{A2}, T_{D1}) > t\}$$

$$\rightarrow E[T] = E[\min(T_{A2}, T_{D1})]$$

Question 3

How to derive $P\{\min(T_{A2}, T_{D1}) > t\}$?

$$P\{\min(T_{A2}, T_{D1}) > t\} = P\{T_{A2} > t, T_{D1} > t\}$$

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How to derive $P\{\min(T_{A2}, T_{D1}) > t\}$?

$$\begin{aligned} P\{\min(T_{A2}, T_{D1}) > t\} &= P\{T_{A2} > t, T_{D1} > t\} \\ &= P\{T_{A2} > t\} P\{T_{D1} > t\} \\ &= (1 - P\{T_{A2} \leq t\})(1 - P\{T_{D1} \leq t\}) \\ &= e^{-\lambda_{A2}t} e^{-\lambda_{D1}t} \\ &= e^{-(\lambda_{A2} + \lambda_{D1})t} \end{aligned}$$

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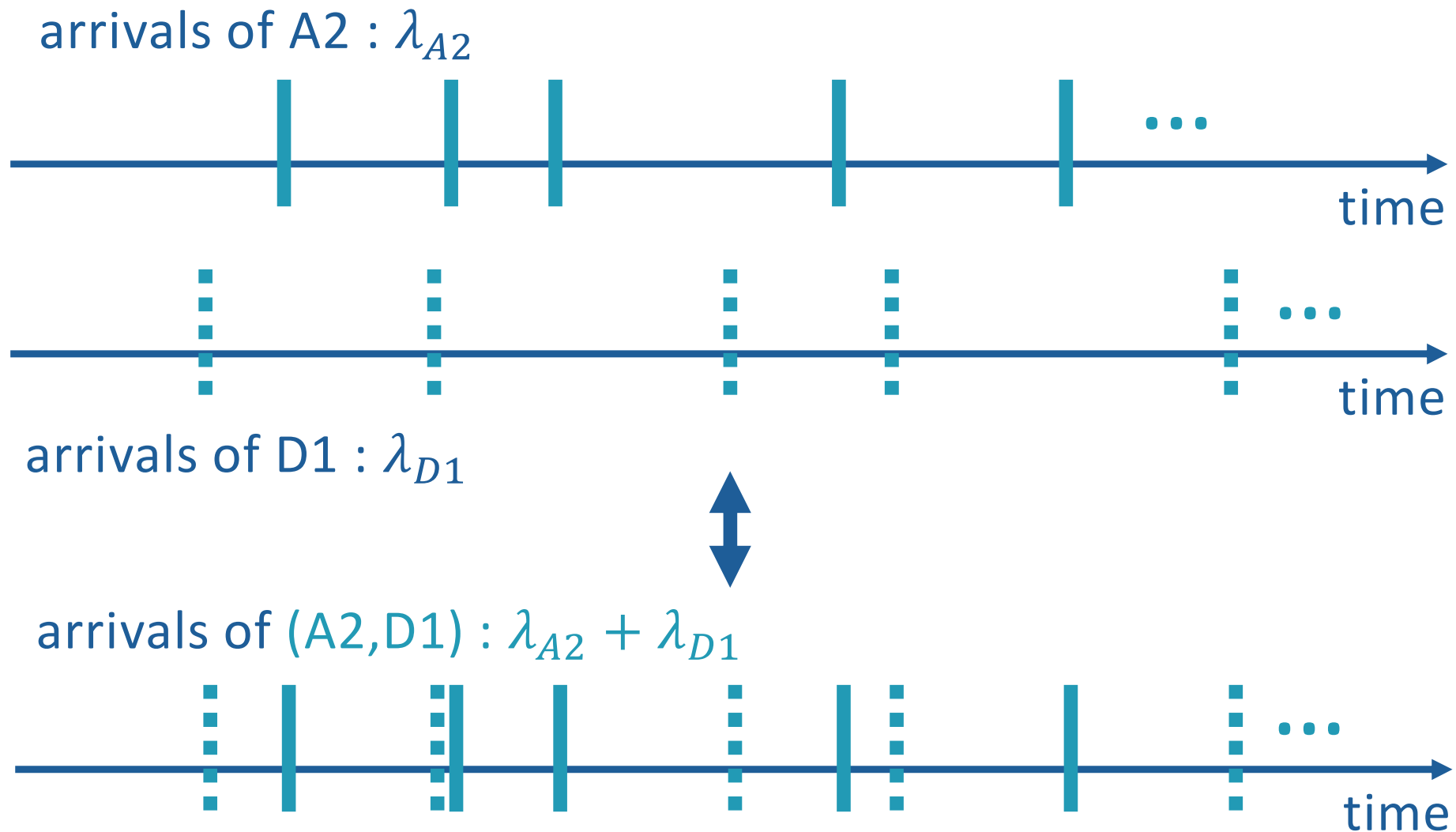
We can notice that $P\{\min(T_{A2}, T_{D1}) > t\} = P\{T > t\}$.

T is the inter-arrival time of a Poisson process with rate $\lambda_{A2} + \lambda_{D1}$.

→ Combination (superposition) of Poisson processes is still a Poisson process.

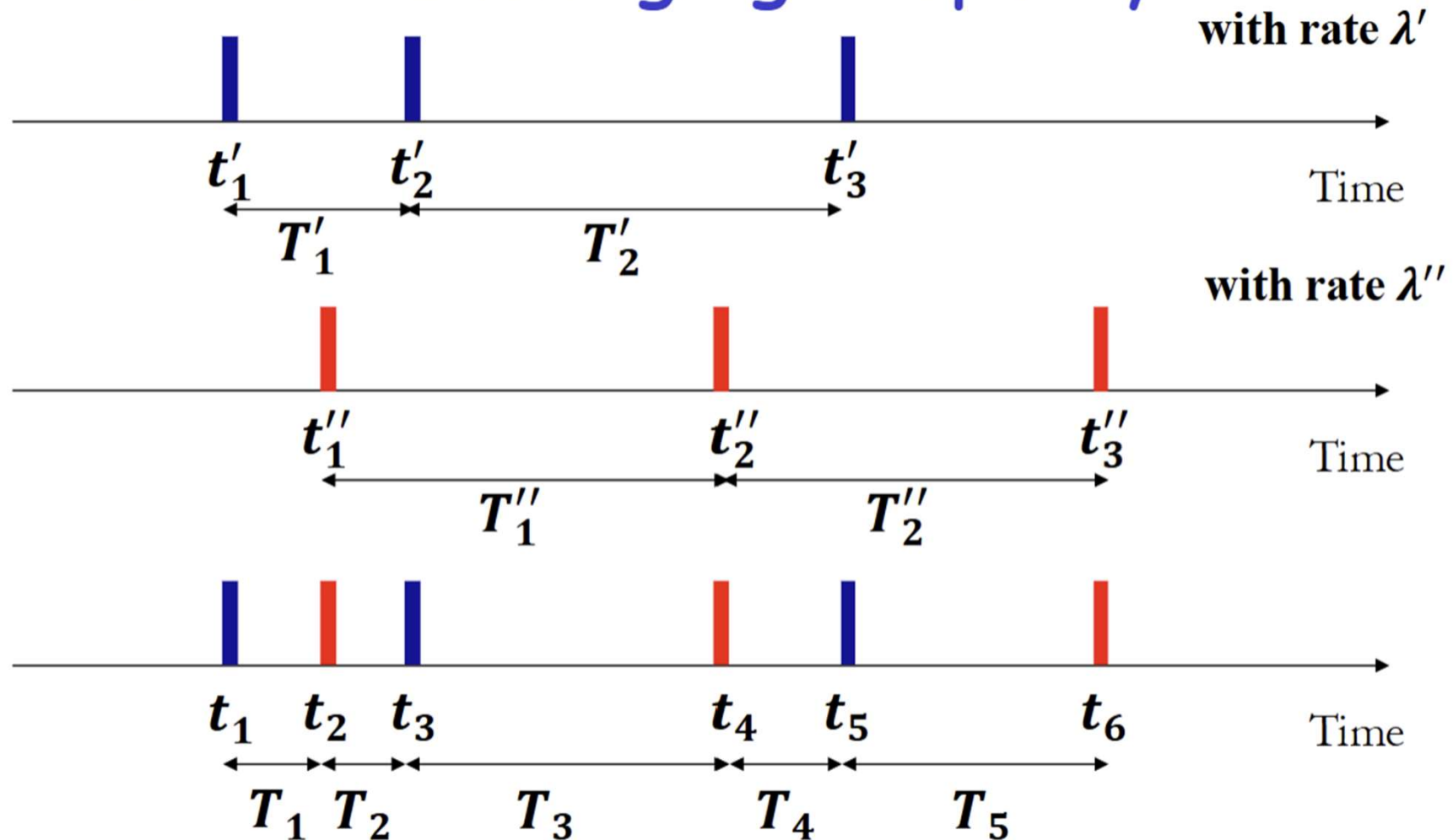
Question 3

Combination of Poisson processes is still a Poisson process.



Question 3

Poisson and Merging Property



□ T is exponential with rate $\lambda = \lambda' + \lambda''$

Question 3

Average waiting time for the next bus

If T follows an exponential distribution with rate λ , the mean is given by $E[T] = 1/\lambda$.

$$\rightarrow E[\min(T_{A2}, T_{D1})] = 1/(\lambda_{A2} + \lambda_{D1}).$$

Question 3

Suppose the arrivals of A2 buses and D1 buses follow Poisson processes with rate λ_{A2} and λ_{D1} . Suppose you want to go to the Central Library from SoC and can take either A2 or D1.

~~Unfortunately, when you arrived at the bus stop, you see an A2 bus and a D1 bus have just departed.~~

How long is your average waiting time for you to get on your bus?

Now will the result be different?

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Now will the result be different?

Recall the memoryless property:

$$P\{T_{wait} > t | T > s\} = P\{T > s + t | T > s\} = P\{T > t\}$$

→ ‘have just departed’ = ‘have departed for $s = 0$ ’

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Now will the result be different?

Recall the **memoryless property**:

→ 'have just departed' = 'have departed for $s = 0$ '

→ The value of s does not affect the final result.

Question 4

Suppose we know that the probability that an M/M/1 system having i customers equals $\pi_i = \rho^i(1 - \rho)$. Derive the expected number of customers in the system. How about the expected number of customers in the queue?

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- L : The number of customers in the system. $\rightarrow$ r.v.
- Q : The number of customers in the queue. $\rightarrow$ r.v.

How to derive the expectation of a random variable?

Question 4

How to derive the expectation of a random variable X ?

- Finite

$$E[X] = x_1 P\{X = x_1\} + \cdots + x_k P\{X = x_k\}$$

- Countably infinite

$$E[X] = \sum_{i=1}^{\infty} x_i P\{X = x_i\}$$

- Absolutely continuous

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

where $f(x)$ is the probability density function.

Question 4

How to derive the expectation of a random variable X ?

Expectation represents the long-run average value of repetitions of the experiment.

e.g., The expected outcome of a roll of a fair six-sided die is 3.5, derived from

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5$$

Question 4

How to derive the expectation of random variable L ?

(Countably infinite case)

$$P\{L = i\} = \pi_i = \rho^i(1 - \rho)$$

$$\begin{aligned} E[L] &= \lim_{n \rightarrow \infty} \sum_{i=0}^n i \cdot P\{L = i\} = \lim_{n \rightarrow \infty} \sum_{i=0}^n i \cdot \rho^i(1 - \rho) \\ &= (1 - \rho) \lim_{n \rightarrow \infty} \sum_{i=1}^n i \cdot \rho^i \end{aligned}$$

Let

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n i \cdot \rho^i \rightarrow E[L] = (1 - \rho) \cdot A$$

Question 4

How to derive the expectation of random variable L ?

$$A = \rho + 2\rho^2 + 3\rho^3 + 4\rho^4 + \dots$$

$$\rho \cdot A = \rho^2 + 2\rho^3 + 3\rho^4 + \dots$$

$$E[L] = (1 - \rho) \cdot A = \rho + \rho^2 + \rho^3 + \rho^4 + \dots - n \rho^{n+1}$$

$$\lim_{n \rightarrow \infty} n \rho^{n+1} = 0$$

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$$E(L) = \rho + \rho^2 + \rho^3 + \rho^4 + \dots$$

$$\rho \cdot E(L) = \rho^2 + \rho^3 + \rho^4 + \dots$$

$$(1 - \rho) \cdot E(L) = \rho - \rho^{n+1}$$

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$$E[L] = \lim_{n \rightarrow \infty} \frac{\rho(1 - \rho^n)}{1 - \rho}$$

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How to derive the expectation of random variable L ?

$$\begin{aligned} A &= \rho + 2\rho^2 + 3\rho^3 + 4\rho^4 + \dots \\ \rho \cdot A &= \rho^2 + 2\rho^3 + 3\rho^4 + \dots \end{aligned}$$

$$\begin{aligned} E[L] &= (1 - \rho) \cdot A = \rho + \rho^2 + \rho^3 + \rho^4 + \dots \\ &= \lim_{n \rightarrow \infty} \frac{\rho(1 - \rho^n)}{1 - \rho} \quad \lim_{n \rightarrow \infty} 1 - \rho^n = 1 \\ &= \frac{\rho}{1 - \rho} \end{aligned}$$

$$\rightarrow E[L] = (1 - \rho) \cdot A = \frac{\rho}{1 - \rho}$$

Question 4

How to derive the expectation of random variable Q ?

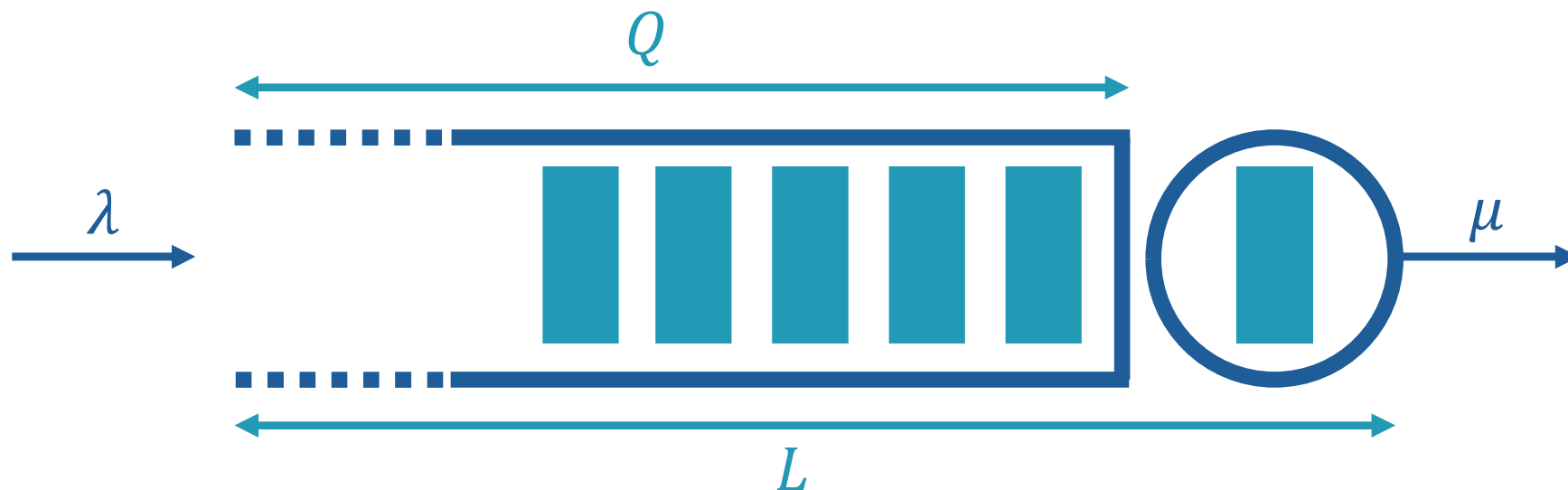
- Relationship between Q and L under M/M/1 model?



Question 4

How to derive the expectation of random variable Q ?

- Relationship between Q and L under M/M/1 model?

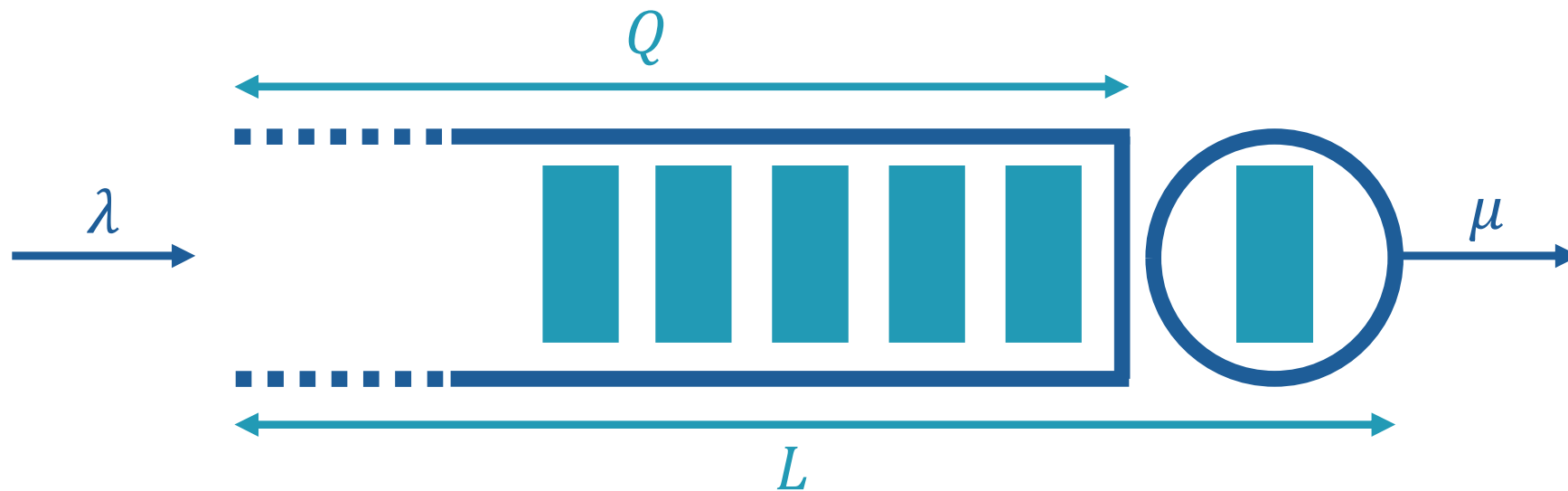


$$E[L] = E[Q] + E[\# \text{ of users being served}]$$

Question 4

How to derive the expectation of random variable Q ?

- Relationship between Q and L under M/M/1 model?



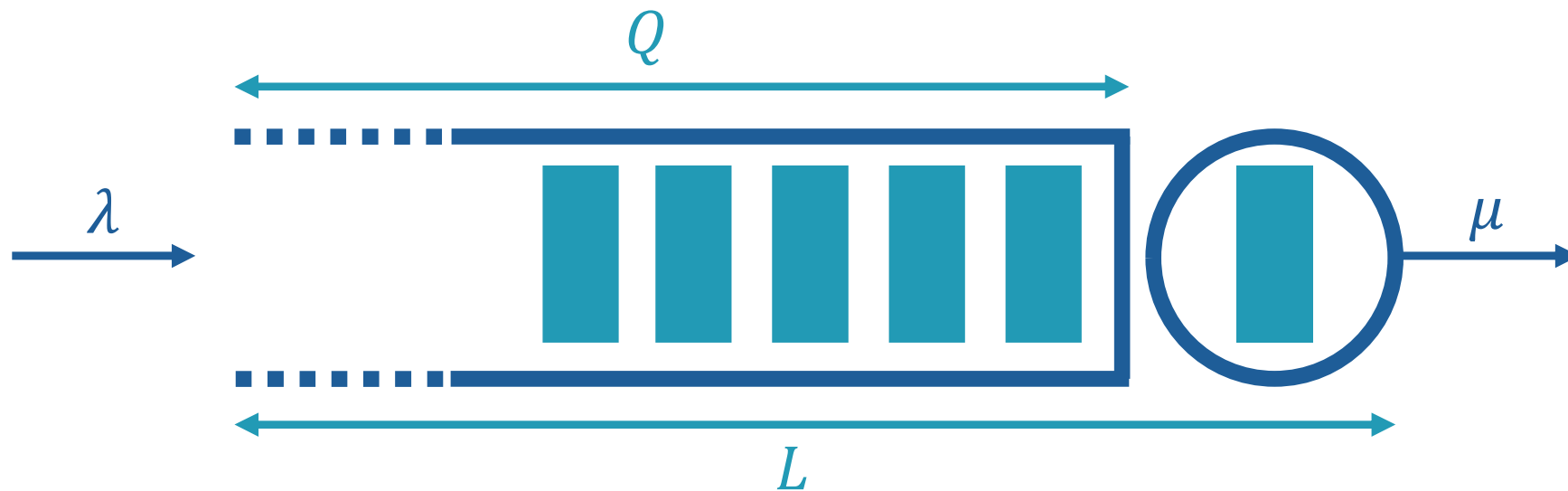
$$E[L] = E[Q] + E[\text{\# of users being served}]$$

$$E[\text{\# of users being served}] = 0(1 - \rho) + 1\rho = \rho$$

Question 4

How to derive the expectation of random variable Q ?

- Relationship between Q and L under M/M/1 model?



$$E[L] = E[Q] + \rho = \frac{\rho}{1 - \rho}$$

$$E[Q] = E[L] - \rho = \frac{\rho^2}{1 - \rho}$$

Thank you.